

Data Analyst

Power BI DAX

MOST ASKED INTERVIEW QUESTIONS

Here are 20 advanced Power BI DAX questions and answers with realworld business scenarios to help you master complex DAX concepts and apply them to business use cases:



1. Running Total of Sales

Scenario: You want to analyze how sales have accumulated month-over month

DAX: RunningTotalSales =

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    FILTER(  
        ALLSELECTED(Sales[Date]),  
        Sales[Date] <= MAX(Sales[Date])  
    )  
)
```

2. Year-over-Year (YoY) Growth

Scenario: Calculate YoY growth in revenue for performance comparison.

DAX:

```
YoYGrowth = VAR CurrentYear = [Total Sales]  
VAR LastYear = CALCULATE([Total Sales],  
    SAMEPERIODLASTYEAR(Sales[Date]))  
RETURN  
DIVIDE(CurrentYear - LastYear, LastYear)
```



3. Sales for the Last N Days

Scenario: Show sales performance over the last 30 days.

DAX:

```
SalesLast30Days = CALCULATE( [Total Sales],  
DATESINPERIOD(Sales[Date],  
MAX(Sales[Date]), -30, DAY)  
)
```

4. Top N Customers by Profit

Scenario: Identify top 5 customers based on profit.

DAX:

```
TopCustomers = TOPN(5, SUMMARIZE  
(Sales,  
Sales[CustomerName],  
"TotalProfit",  
SUM(Sales[Profit])),  
[TotalProfit]  
)
```



5. Dynamic Ranking

Scenario: Rank products based on their total sales.

DAX:

```
ProductRank = RANKX(ALL(Sales[ProductName]),  
[Total Sales], ,  
DESC  
)
```

6. Customer Retention Rate

Scenario: Track how many customers from the previous year returned this year.

DAX:

```
RetentionRate =  
DIVIDE(  
    COUNTROWS(  
        INTERSECT(  
            VALUES(Customers[CustomerID]),  
            CALCULATETABLE(VALUES(Customers[CustomerID]),  
                SAMEPERIODLASTYEAR(Sales[Date])) ) ),  
    CALCULATE(DISTINCTCOUNT(Customers[CustomerID]),  
        SAMEPERIODLASTYEAR(Sales[Date]))  
)  
)
```



7. Previous Month Sales

Scenario: Compare current month to the previous month's sales.

DAX:

```
PreviousMonthSales = CALCULATE([Total Sales],  
PREVIOUSMONTH(Sales[Date])  
)
```

8. Forecast Next Month's Sales

Scenario: Predict next month's sales based on a simple moving average.

DAX:

```
ForecastSales = AVERAGEX(  
    LASTN(3, DATESINPERIOD(Sales[Date],  
        MAX(Sales[Date]), -3, MONTH)),  
    CALCULATE([Total Sales])  
)
```



9. Calculate Profit Margin %

Scenario: Determine how profitable a product is.

DAX:

```
ProfitMargin = DIVIDE(SUM(Sales[Profit]),  
SUM(Sales[SalesAmount])  
)
```

10. Filter by Latest Date

Scenario: Show data only for the most recent transaction date.

DAX:

```
LatestDateSales = CALCULATE(  
    [Total Sales],  
    Sales[Date] = MAX(Sales[Date])  
)
```



11. Sales Target Achievement %

Scenario: Compare actual sales to a predefined target.

DAX:

```
TargetAchieved =  
DIVIDE([Total Sales],  
       [Sales Target]  
)
```

12. Dynamic Measure Based on Selection

Scenario: Switch between sales, profit, or quantity using a slicer.

DAX:

```
SelectedMeasure = SWITCH(  
    SELECTEDVALUE(MeasureSelector[MeasureName]),  
    "Sales", [Total Sales],  
    "Profit", [Total Profit],  
    "Quantity", [Total Quantity]  
)
```



13. Moving Average (3 Months)

Scenario: Smooth out sales trends over 3 months.

DAX:

```
MovingAvg = AVERAGEX(
    DATESINPERIOD(Sales[Date],
        MAX(Sales[Date]), -3, MONTH),
    [Total Sales]
)
```

14. Cumulative Sales by Category

Scenario: Cumulative sales per category.

DAX:

```
CumulativeSalesCategory = CALCULATE(
    [Total Sales],
    FILTER( ALLSELECTED(Sales[Date]),
        Sales[Date] <= MAX(Sales[Date])
    )
)
```



15. Customer Lifetime Value (CLV)

Scenario: Estimate revenue generated by each customer over time.

DAX:

```
CustomerLTV = CALCULATE([Total Sales],  
    ALLEXCEPT(  
        Sales, Sales[CustomerID]  
    )  
)
```

16. Percentage Contribution to Total Sales

Scenario: How much does each product contribute to total sales?

DAX:

```
%Contribution = DIVIDE([Total Sales],  
    CALCULATE([Total Sales],  
        ALL(Sales)  
    )  
)
```



17. Compare Sales to Average Sales

Scenario: Compare each product's sales to the average.

DAX:

```
AboveAverage = IF(
    [Total Sales] > AVERAGE(Sales[SalesAmount]),
    "Above Avg",
    "Below Avg"
)
```

18. Dynamic Time Intelligence Measure

Scenario: Show sales based on user-selected time frame (MTD, QTD, YTD).

DAX:

```
DynamicTimeMeasure = SWITCH(
    SELECTEDVALUE(TimePeriod[Period]),
    "MTD", TOTALMTD([Total Sales],
        Sales[Date]), "QTD",
    TOTALQTD([Total Sales], Sales[Date]),
    "YTD", TOTALYTD(
        [Total Sales], Sales[Date]
    )
)
```



19. Churn Rate

Scenario: Measure how many customers stopped purchasing.

DAX:

```
ChurnRate = 1 - DIVIDE(
    COUNTROWS( INTERSECT( VALUES(Sales[CustomerID]),
        CALCULATETABLE(VALUES(Sales[CustomerID]),
            PREVIOUSYEAR(Sales[Date])) ) ),
    CALCULATE(DISTINCTCOUNT(Sales[CustomerID]),
        PREVIOUSYEAR(Sales[Date])
    )
)
```

20. Dynamic Titles Based on Filters

Scenario: Create report titles that change based on selected region or date.

DAX:

```
DynamicTitle = "Sales Report for " & SELECTEDVALUE(
    Region[RegionName] & " - " & FORMAT(MAX(Sales[Date]),
        "MMMM YYYY"
    )
)
```

